

Exercises

1 Probabilities

Given a Doctor's Patient Record, where each cell $n(x_r, \theta_c)$ represents the number of patients with symptom x_r and disease θ_c :

	θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	θ_7	θ_8	θ_9	θ_{10}	Sum
x_1	8	9	9	5	4	1	1	0	0	0	37
x_2	3	5	8	9	14	10	3	3	0	0	55
x_3	0	1	1	10	16	11	12	7	8	5	71
x_4	0	0	1	0	3	5	10	7	7	4	37
Sum	11	15	19	24	37	27	26	17	15	9	200

Table 1: Joint observations of symptoms x_r (rows) and diseases θ_c (columns).

Questions from the doctor:

- **Q1:** What is the joint probability $P(x_3, \theta_2)$ that a patient has the symptom x_3 and the disease θ_2 ?
- **Q2:** What is the probability $P(x_3)$ that a patient has the symptom x_3 ?
- **Q3:** What is the probability $P(\theta_2)$ that a patient has the disease θ_2 ?
- **Q4:** What is the conditional probability $P(x_3|\theta_2)$ that a patient has the symptom x_3 given that they have the disease θ_2 ?
- **Q5:** What is the conditional probability $P(\theta_2|x_3)$ that a patient has the disease θ_2 given that they have the symptom x_3 ?

Solution

- We use random variables to model the patient record:
 - Let the discrete random variable X represent the row number, i.e., one of the four symptoms.
 - The range of X is: $R_X = \{x_1, x_2, x_3, x_4\}$.
 - Let Θ represent the column number, i.e., one of the ten diseases.
 - The range of Θ is: $R_\Theta = \{\theta_1, \theta_2, \dots, \theta_{10}\}$.

	θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	θ_7	θ_8	θ_9	θ_{10}	$P(X)$
x_1	0.04	0.045	0.045	0.025	0.02	0.005	0.005	0	0	0	0.185
x_2	0.015	0.025	0.04	0.045	0.07	0.05	0.015	0.015	0	0	0.275
x_3	0	0.005	0.005	0.05	0.08	0.055	0.06	0.035	0.04	0.025	0.355
x_4	0	0	0.005	0	0.015	0.025	0.05	0.035	0.035	0.02	0.185
$P(\Theta)$	0.055	0.075	0.095	0.12	0.185	0.135	0.13	0.085	0.075	0.045	1

Table 2: Joint PMF of two discrete random variables X and Θ and their marginal PMFs. The marginal PMFs $P(X)$ and $P(\Theta)$ are also known as the **marginal likelihood** (of symptoms) and the **prior distribution** (of disease), respectively.

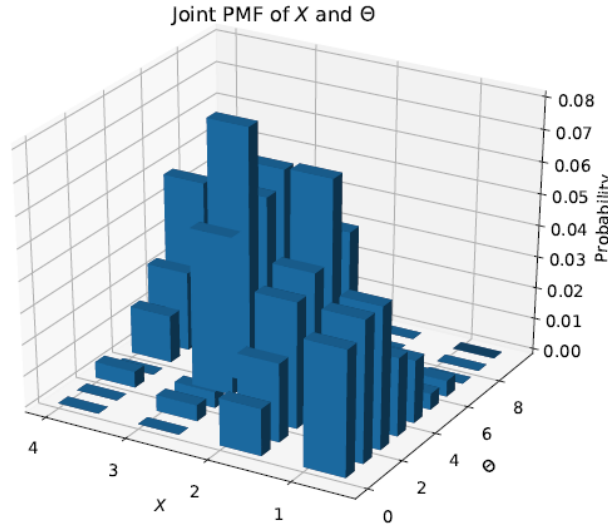


Figure 1: Joint PMF of X and Θ

- We can now calculate the joint probability and obtain the joint probability mass function (PMF) of X and Θ .

Answers to the Doctor's questions:

Q1: What is the joint probability $P(x_3, \theta_2)$ that a patient has the symptom x_3 and the disease θ_2 ? **A1:** $P(x_3, \theta_2) = 0.005$

Q2: What is the probability $P(x_3)$ that a patient has the symptom x_3 ? **A2:** $P(x_3) = 0.355$

Q3: What is the probability $P(\theta_2)$ that a patient has the disease θ_2 ? **A3:** $P(\theta_2) = 0.075$

Q4: What is the conditional probability $P(x_3|\theta_2)$ that a patient has the

symptom x_3 given that they have the disease θ_2 ?

$$P(x_3|\theta_2) = \frac{P(x_3, \theta_2)}{P(\theta_2)} = \frac{0.005}{0.075} = \frac{1}{15}$$

Q5: What is the conditional probability $P(\theta_2|x_3)$ that a patient has the disease θ_2 given that they have the symptom x_3 ?

$$P(\theta_2|x_3) = \frac{P(x_3, \theta_2)}{P(x_3)} = \frac{0.005}{0.355} = \frac{1}{71}$$

2 Maximum Likelihood

1. We flipped a coin 100 times. Given that there were 55 heads, use maximum likelihood estimation to find the probability p of heads on a single toss.
2. Let X be independent and identically distributed (i.i.d.) $\text{Poisson}(\lambda)$ distributed. Find the maximum likelihood estimator for λ , $\hat{\lambda}$. Calculate an estimate using this estimator when:

$$x_1 = 1, x_2 = 2, x_3 = 5, x_4 = 3.$$

Solutions

1. Coin Toss Experiment

Step 1: Likelihood Function

Since coin flips follow a binomial distribution:

$$P(55 \text{ heads}|p) = \binom{100}{55} p^{55} (1-p)^{45}$$

Step 2: Log-Likelihood Function

$$\log P(55 \text{ heads}|p) = \log \binom{100}{55} + 55 \log p + 45 \log(1-p)$$

Step 3: Differentiate and Solve for p

$$\frac{d}{dp} (55 \log p + 45 \log(1-p)) = \frac{55}{p} - \frac{45}{1-p} = 0$$

Solving for p :

$$55(1-p) = 45p$$

$$55 = 100p$$

$$\hat{p} = 0.55$$

2. MLE for Poisson Distribution

Step 1: Poisson Probability Mass Function (PMF)

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

Step 2: Log-Likelihood Function

$$L(\lambda|x_1, \dots, x_n) = \prod_{j=1}^n \frac{\lambda^{x_j} e^{-\lambda}}{x_j!}$$

$$\log L(\lambda) = \sum_{j=1}^n x_j \log \lambda - \lambda - \sum_{j=1}^n \log x_j!$$

Step 3: Differentiate and Solve for λ

$$\frac{d}{d\lambda} \left(\sum_{j=1}^n x_j \log \lambda - \lambda \right) = \frac{\sum x_j}{\lambda} - n = 0$$

$$\hat{\lambda} = \frac{1}{n} \sum x_j$$

For the given data:

$$\hat{\lambda} = \frac{1 + 2 + 5 + 3}{4} = 2.75$$

3 Logistic Regression

A company wants to predict whether a customer will purchase a product based on their income and age. The logistic regression model is given by:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2)}}$$

where:

- $Y = 1$ if the customer buys the product, $Y = 0$ otherwise.
- X_1 = Age of the customer.
- X_2 = Monthly income of the customer (in £1000s).
- $\beta_0 = -3$, $\beta_1 = 0.05$, $\beta_2 = 0.1$.

Tasks:

1. Compute the probability that a customer $\mathbf{x}$ aged 30 with a monthly income of £5000 will purchase the product.

2. Interpret the meaning of $\beta_1 = 0.05$ in the context of this problem.
3. If a customer's probability of purchasing is 0.7, should the company predict that they will buy the product? Justify your answer.
4. Briefly explain one advantage and one limitation of using logistic regression for this problem.

Solution

1. Computing probability:

$$\mathbf{x} = (1, 30, 5); \quad \beta = (-3, 0.05, 0.1).$$

$$\beta^T \mathbf{x} = (-3) + (0.05 \times 30) + (0.1 \times 5) = -1$$

$$P(Y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(-1)}} = \frac{1}{1 + e^1} = \frac{1}{3.718} \approx 0.269$$

2. Interpretation of β_1 : A one-year increase in age increases the log-odds of purchasing by 0.05, meaning age has a positive effect on purchase probability.

3. Prediction for $P(Y = 1|\mathbf{x}) = 0.7$: Since $0.7 > 0.5$, the model predicts that the customer will buy the product.

4. Advantage and limitation:

- **Advantage:** Interpretability – coefficients show how features affect probability.
- **Limitation:** Assumes a linear relationship in log-odds, which may not always hold.

4 Naïve Bayes - Disease Diagnosis

A medical test is used to diagnose a rare disease based on two symptoms: Fever and Cough. The model uses a Naïve Bayes classifier.

The probabilities of symptoms appearing given the disease status are:

Symptom	$P(\text{symptom} \text{disease})$	$P(\text{symptom} \text{no disease})$
Fever	0.8	0.2
Cough	0.6	0.3

Additionally, we know that only 5% of the population has the disease (i.e., $P(\text{disease}) = 0.05$).

Tasks:

1. Compute the probability that a patient with both Fever and Cough has the disease, and classify the patient as having or not having the disease.
2. What is the 'naive Bayes' assumption and why is it important in Naïve Bayes classification?

Solution

1. Computing probability:

$$P(\text{Fever, Cough}|\text{Disease})P(\text{Disease}) = (0.8 \times 0.6) \times 0.05$$

$$= 0.48 \times 0.05 = 0.024$$

For no disease:

$$P(\text{Fever, Cough}|\text{No Disease})P(\text{No Disease}) = (0.2 \times 0.3) \times 0.95$$

$$= 0.06 \times 0.95 = 0.057$$

Now, compute the posterior probability:

$$P(\text{Disease}|\text{Symptoms}) = \frac{0.024}{0.024 + 0.057} = \frac{0.024}{0.081} \approx 0.296$$

Since $0.296 < 0.5$, the model does **not** classify the patient as having the disease.

Note: If the question only asked for the decision (Disease or No Disease) and not the probability, then it would have been enough to compute and compare the numerators of $P(\text{Disease}|\text{Symptoms})$ and $P(\text{No Disease}|\text{Symptoms})$. Always read the question carefully.

2. On the naive Bayes (conditional independence) assumption:

The assumption simplifies computation by treating symptoms as independent given the disease. This makes the model fast and scalable, though symptoms in reality may be correlated.

Gradient of the Logistic Regression Cost Function $J(\theta)$

The cost function for logistic regression is defined as:

$$J(\theta) = - \sum_i [y_i \log \sigma(x_i) + (1 - y_i) \log(1 - \sigma(x_i))]$$

where:

$$\sigma(x_i) = \frac{1}{1 + \exp(-\theta^\top x_i)}$$

is the sigmoid function.

Gradient of $J(\theta)$ with respect to θ_k

To compute the gradient, we take the partial derivative of $J(\theta)$ with respect to a parameter θ_k . For a single data point i , the cost function contribution is:

$$J_i(\theta) = - [y_i \log \sigma(x_i) + (1 - y_i) \log(1 - \sigma(x_i))]$$

Step 1: Derivative of $\log \sigma(x_i)$

Using the chain rule:

$$\frac{\partial}{\partial \theta_k} \log \sigma(x_i) = \frac{1}{\sigma(x_i)} \cdot \frac{\partial \sigma(x_i)}{\partial \theta_k}$$

The derivative of the sigmoid function is:

$$\frac{\partial \sigma(x_i)}{\partial \theta_k} = \sigma(x_i)(1 - \sigma(x_i)) \cdot x_{ik}$$

Thus:

$$\frac{\partial}{\partial \theta_k} \log \sigma(x_i) = \frac{\sigma(x_i)(1 - \sigma(x_i)) \cdot x_{ik}}{\sigma(x_i)} = (1 - \sigma(x_i)) \cdot x_{ik}$$

Step 2: Derivative of $\log(1 - \sigma(x_i))$

Similarly:

$$\frac{\partial}{\partial \theta_k} \log(1 - \sigma(x_i)) = \frac{1}{1 - \sigma(x_i)} \cdot \frac{\partial(1 - \sigma(x_i))}{\partial \theta_k}$$

Since:

$$\frac{\partial(1 - \sigma(x_i))}{\partial \theta_k} = -\sigma(x_i)(1 - \sigma(x_i)) \cdot x_{ik}$$

we get:

$$\frac{\partial}{\partial \theta_k} \log(1 - \sigma(x_i)) = -\frac{\sigma(x_i)(1 - \sigma(x_i)) \cdot x_{ik}}{1 - \sigma(x_i)} = -\sigma(x_i) \cdot x_{ik}$$

Step 3: Combining Terms

The derivative of the cost function for a single data point is:

$$\frac{\partial J_i(\theta)}{\partial \theta_k} = -y_i(1 - \sigma(x_i)) \cdot x_{ik} - (1 - y_i)(-\sigma(x_i) \cdot x_{ik})$$

Simplify:

$$\frac{\partial J_i(\theta)}{\partial \theta_k} = (\sigma(x_i) - y_i) \cdot x_{ik}$$

Step 4: Summing Over All Data Points

The total gradient is the sum over all data points:

$$\frac{\partial J(\theta)}{\partial \theta_k} = \sum_i (\sigma(x_i) - y_i) \cdot x_{ik}$$

Final Gradient Expression

In vectorized form, the gradient can be written as:

$$\nabla J(\theta) = X^\top (\sigma(X\theta) - y)$$

where:

- X is the design matrix ($N \times K$).
- $\sigma(X\theta)$ is the vector of predicted probabilities.
- y is the vector of true labels.

This gradient is used in gradient-based optimization methods, such as Gradient Descent, to minimize $J(\theta)$.

5 Larger models and recent trends (a selection)

Extra reading: Artificial Neural Networks and trends in AI Artificial Neural Networks

I recommend this old tutorial if you are interested in artificial neural networks:

”Artificial neural networks: a tutorial”, by A.K. Jain; Jianchang Mao; K.M. Mohiuddin, IEEE Computer, Volume: 29, Issue: 3, March 1996.
<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=485891>

Trends in AI (e.g., LLMs) - AI and jobs

”Will AI Take Your Job? Maybe Not Just Yet, One Study Says”, by Billy Perrigo, Time Magazine, 2024 (5 minutes read)
<https://time.com/6565026/ai-job-replacement-mit-study/>

The paper mentioned in the Time Magazine article:
”Beyond AI Exposure: Which Tasks are Cost-Effective to Automate with Computer Vision?”
https://futuretech-site.s3.us-east-2.amazonaws.com/2024-01-18+Beyond_AI_Exposure.pdf

RAG (Retrieval Augmented Generation) Current LLMs tend to be inconsistent and unreliable, e.g., hallucinations. To make LLMs more reliable, Meta proposed RAG in 2020:

”Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”
<https://arxiv.org/abs/2005.11401v4>

A summary of RAG is available here:
<https://research.ibm.com/blog/retrieval-augmented-generation-RAG>

Small language models

Even Microsoft successfully bet on LLMs (OpenAI’s chatGPT) and became the Second Company Ever To Top \$3 Trillion Valuation.
<https://www.forbes.com/sites/dereksaul/2024/01/24/microsoft-becomes-second-company-ever-to-top-3-trillion-valuation/?sh=6053083c40ac>

They recently discovered the surprising power of small language models.
<https://www.microsoft.com/en-us/research/blog/phi-2-the-surprising-power-of-small-language-models/>